

Jesse Ramirez

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EDUCATION

University of California, Berkeley

Expected Spring 2026

Bachelor of Science in Mechanical Engineering, Minor in Aerospace

GPA 3.2

- Relevant Coursework: Thermodynamics, Heat Transfer, Fluid Mechanics, Flight Mechanics, Dynamics & Controls, Mechanics of Materials, Solid Mechanics, Composite Materials, Combustions, IoT, 3D Modeling & Manufacturing

TECHNICAL SKILLS

Software: SolidWorks, Siemens NX, Fusion360, RPA, OnShape, Revit, Ansys, MATLAB, Python, Altium Designer

Manufacturing: 3D printing, CNC, Milling, Lathe, Water Jet, Laser Cutting, Soldering, Welding

TECHNICAL EXPERIENCE

Southland Industries

May 2025 - August 2025

Design Engineering Intern

Union City, CA

- Created a lab matrix calculator for the Google Geneva Data Center to determine water and air cooling capacities for various lab equipment, enabling Google to assess each space's power consumption and corresponding cooling demands
- Verified cooling capacity for fan coil units in laboratory spaces, cross-checking design intent with actual performance criteria to ensure adequate capacity, energy efficiency, and compliance with project specifications
- Executed the complete design process for a tenant improvement space of 4,250 sqft, performing detailed load calculations, developing HVAC layouts, and designing mechanical piping systems

Space Technology and Rocketry (STAR)

September 2023 - Present

Propulsion Lead

Berkeley, CA

- Designed a regeneratively cooled LOX/RP-1 thrust chamber 1050 lbf partnering with 3D Systems to apply DFM for metal DMP; iterating cooling-channel geometry, minimum wall/feature thickness, and post-processing planning
- Led re-hotfire integration on our launch vehicle owning avionics/flight hardware prep including wiring diagrams, harness routing/strain relief, soldered terminations, load-cell integration, and solenoid actuation checkout
- Designed the full P&ID, CAD model, and physical assembly for the ground and fluid systems of a pressure-fed, integrating propellant lines, valves, and instrumentation for testing operations

Undergraduate Research at the Berkeley Fire Lab

February 2025 - Present

Student Researcher

Berkeley, CA

- Conducted research for NASA's SoFIE-MIST investigation aboard the International Space Station, analyzing oxygen concentration, pressure variation, and preheating by performing surface heat flux tests to validate flammability models and improve fire safety in microgravity
- Led machining and test-fixture fabrication for burn testing rig and supporting hardware, and lathe-machined material samples down to 3.20 mm to improve thermal characterization during experiments

Undergraduate Research at the Berkeley Nanotechnology Lab

February 2025 - August 2025

Student Researcher

Berkeley, CA

- Assisted in research on carbon nanotube-based biosensor platforms, energy solutions, and environmental sensing
- Read schematics and assembled PCB circuitry for multiple sub-teams, supporting integration and test of experimental nanotechnology systems; performed SMD placement and reflow-oven soldering on components down to 0402 package size

NASA L'Space Mission Concept Academy

January 2024 - May 2024

Mechanical and Thermal Engineer

Berkeley, CA

- Led the design of a Martian reconnaissance rover focused on characterizing optimal water feedstock locations
- Validated trade studies for mechanical and thermal subsystems to guide component selection, balancing performance, weight, and reliability under simulated Martian conditions
- Performed recovery and redundancy analysis on the chassis, suspension, and thermal control systems, implementing backup solutions to ensure mission continuity in case of component failure

LEADERSHIP EXPERIENCE

Recreational Sports Facility (RSF)

July 2022 - August 2025

Fitstaff Student Lead

Berkeley, CA

- Led the staff hiring process with RSF management, assessing candidate qualifications to increase recruitment efficiency
- Addressed overcapacity challenges through effective management strategies, reducing room capacity from 115% to 80%

PROJECTS

Bio-Inspired FDM Composite Heat-Shield Tiles | ANSYS, MATLAB

Spring 2025

- Performed transient heat transfer simulations in ANSYS and validated results through physical testing on carbon-nylon and nylon samples, analyzing deformation and heat dissipation across bio-inspired geometries under high thermal loads.
- Developed a thermocouple-based DAQ system for real-time temperature monitoring, enabling accurate detection of material failure thresholds and correlation with simulated thermal gradients.

Rocket Engine Design Automation | Python

June 2024 - Present

- Developed an automated Python tool that integrates performance, cooling, and flow models to iterate rocket engine designs optimizing O/F ratio, mass flow rate, and chamber pressure to meet thrust and thermal constraints
- Implemented a constrained optimization model using the interior-point method to size high-aspect-ratio cooling channels